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1 % This function initializes the parameters for the model and sets initial
2 % values for the variables.
3 function [x0, parsG, Init, low, hi] = load_global(mdata,data)
4
5 global DIFF_INC ODE_TOL
6
7 Qtot = data.Qtot;
8
9 % Flows (related to subject)
10 Qmval = Qtot;
11 Qaval = Qtot;
12 Qsys = Qtot;
13 Qtval = Qtot;
14 Qpval = Qtot;
15 Qpul = Qtot;
16
17 PsaS = data.PsaS;
18 PsaD = data.PsaD;
19 PsaM = PsaS/3 + 2*PsaD/3;
20
21 PpaS = data.PpaS;
22 PpaD = data.PpaD;
23 PpaM = PpaS/3 + 2*PpaD/3;
24
25 Psv = data.Psv;
26 Ppv = data.Ppv;
27
28 %Resistances (Ohm's law)
29 Rpul = (PpaD - Ppv)/Qpul;
30 Rmval = 0.001; % Mitral valve resistance
31 Raval = 0.001; % Arterial valve resistance
32
33 Rsys = (PsaD - Psv)/Qsys;
34 Rtval = 0.001;
35 Rpval = 0.001;
36
37 % Total Volumes (Beneken and deWit)
38
39 VtotA= 0.20*data.TotalVol; % Arterial volume (20%)
40 VtotV= 0.80*data.TotalVol; % Venous volume (80%)
41
42 % Peskin & Hoppensteadt pg 8 (21 on pdf)
43 VpvT = VtotV*0.80; % Upper body arterial volume
44 VsaT = VtotA*0.22; % Lower body arterial volume
45 VsvT = VtotV*0.78; % Upper body venous volume
46 VpaT = VtotA*0.20; % Lower body venous volume
47
48 % Stressed Volumes – used Beneken and deWit for percentages
49 VpvS = VpvT*.075;
50 VsaS = VsaT*.27;
51 VsvS = VsvT*.105;
52 VpaS = VpaT*.58;
53
54 % Unstressed Volumes
55 VpvU = VpvT - VpvS;

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56 VsaU = VsaT - VsaS;
57 VsvU = VsvT - VsvS;
58 VpaU = VpaT - VpaS;
59
60 VlvU = data.VlvU;
61 VrvU = data.VlrU;
62
63 parsG = [VpvU VlvU VsaU VsvU VrvU VpaU];
64
65 % Compliances, stressed volume percentages from Beneken are weighted averages
66 Cpv = VpvS/Ppv; % pulmonary veins compliance
67 Csa = VsaS/PsaS; % systemic arteries compliance
68 Csv = VsvS/Psv; % Lower body arterial compliance
69 Cpa = VpaS/PpaS; % Lower body venous compliance
70
71 VlvM = data.VlvM; % Max volume left ventricle (from literature)
72 VlvM = data.VlvM; % Minimum volume left ventricle (from literature)
73
74 VrvM = data.VrvM; % Max volume right ventricle (from literature)
75 VrvM = data.VrvM; % Minimum volume right ventricle (from literature)
76
77 ElvM = Psv/VlvM; % Minimum cardiac elastance - left heart
78 ElvM = PsaS/VlvM; % Maximum cardiac elastance - left heart
79
80 Ervm = Psv/VrvM; % minimum cardiac elastance - right heart
81 Ervm = PsaS/VrvM; % max cardiac elastance - right heart
82
83 TC = data.TC; % Time for maximum contractility relative to the lenght of
the heart beat
84 TR = data.TR; % Time for relaxation relative to the lenght of the heart beat
85
86
87 Init = [VpvT VlvM VsaT VsvT VrvM VpaT]; % Steady state initial conditions
88
89 % Parameter vector
90 x0 = [Rpul Rmval Raval Rsys Rtval Rpval ... %1-6
91 Cpv Csa Csv Cvpa ... %7-10
92 ElvM Elvm Ervm Ervm TC TR]'; %11-16
93
94 x0 = log(x0);
95
96 % MA432 Required for optimization
97 low = x0 - log(4);
98 hi = x0 + log(4);
99

```